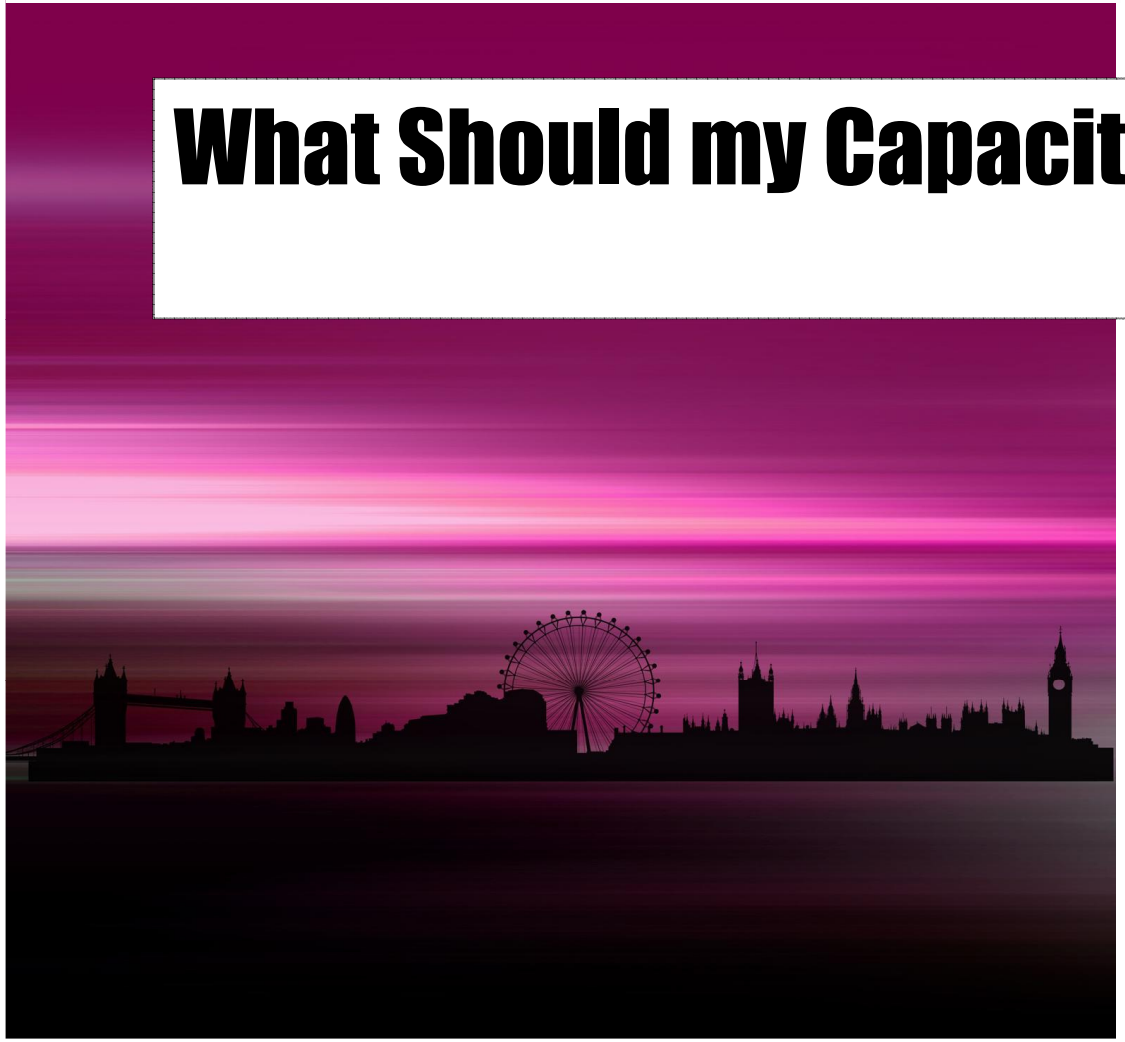
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What Should my Capacity Plan Contain?



Danny Quilton, COO,
Capacitas

Synopsis

Many organisations are now adopting Capacity Management processes. The key deliverable of the Capacity Management process is the capacity plan. For those new to Capacity Management, the question often posed is 'What information should my capacity plan contain?' This presentation offers practical assistance to those looking to deliver a Capacity Plan using real world examples from numerous client engagements. The challenges associated with delivering a capacity plan and inertia around capacity plan findings are discussed. Guidance is also provided on the frequency of capacity plan releases and effective communication methods for delivering the key messages of the plan to senior management.

Agenda

- ITIL v3 definition
- Capacity plan contents – Capacitas's guidance
- Conveying the messages of the capacity plan
- Capacity plan frequency
- Summary

ITIL v3 Capacity Management

- **Objective**

  "The prime objective is to produce a plan that documents the current levels of resource utilisation and service performance, and after consideration of the business strategy and plans, forecasts the future requirements for resource to support IT services that underpin business activities. The plan should indicate clearly any assumptions made. It should also include any recommendations quantified in terms of resource required, cost, benefits, impact etc."

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Contents of a Capacity Plan

- Introduction
- Management Summary
- Scope
- Business Capacity Management
- Service Capacity Management
- Component Capacity Management
- Capacity Plan Scenarios
- Recommendations
- Costs
- Action Log
- Plan vs. Actual
- Options for Service Improvement
- Appendices



Scope

- **It's critical to clearly define the scope**

Period of plan	• Technology refresh typically occurs every 3 years
Components in scope	• Complex technical architectures require a limiting scope • Components may have sub-components • Employ a risk assessment
Interfaces between IT Services	• Which capacity plan do these reside in?

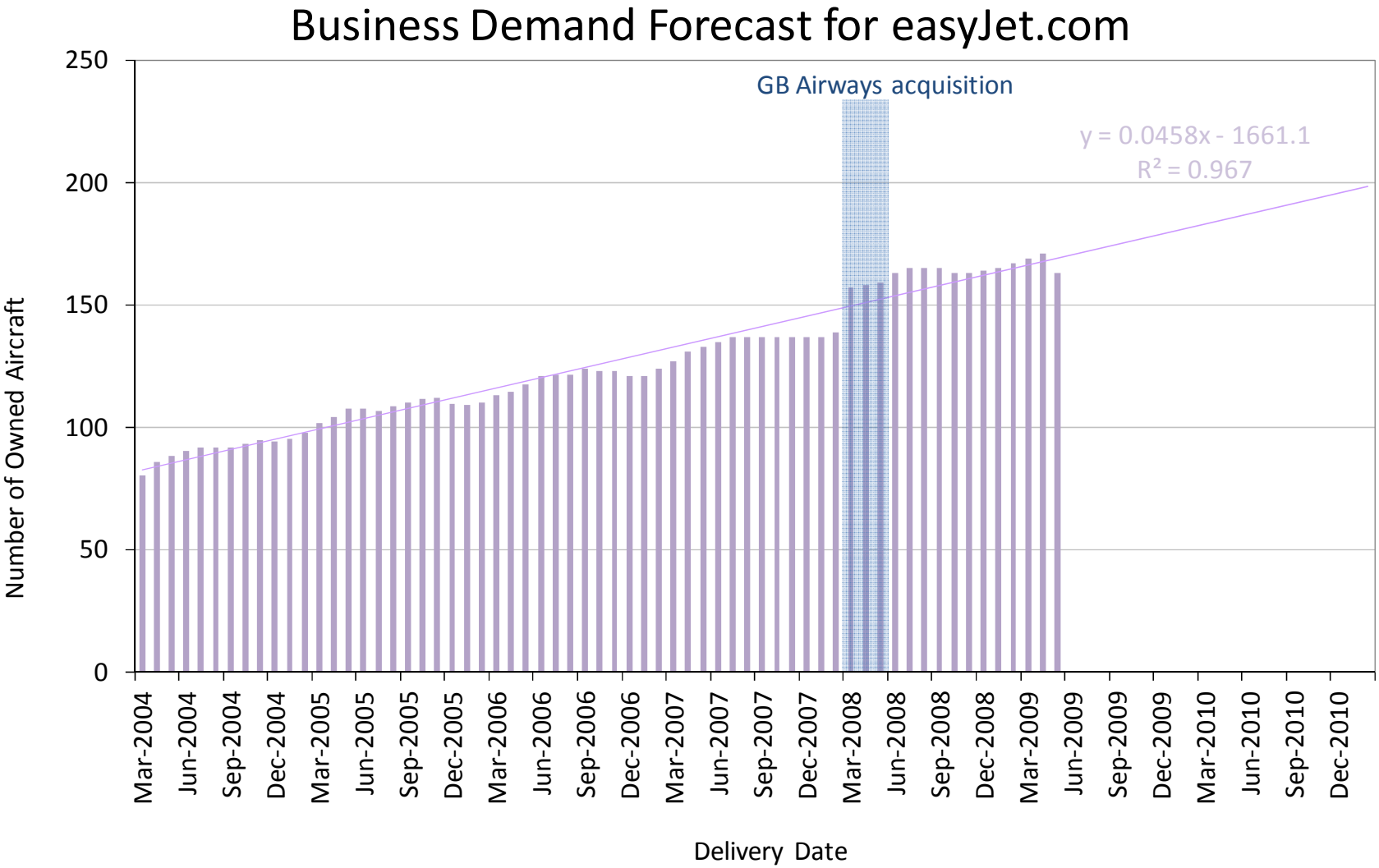


Business Capacity Management

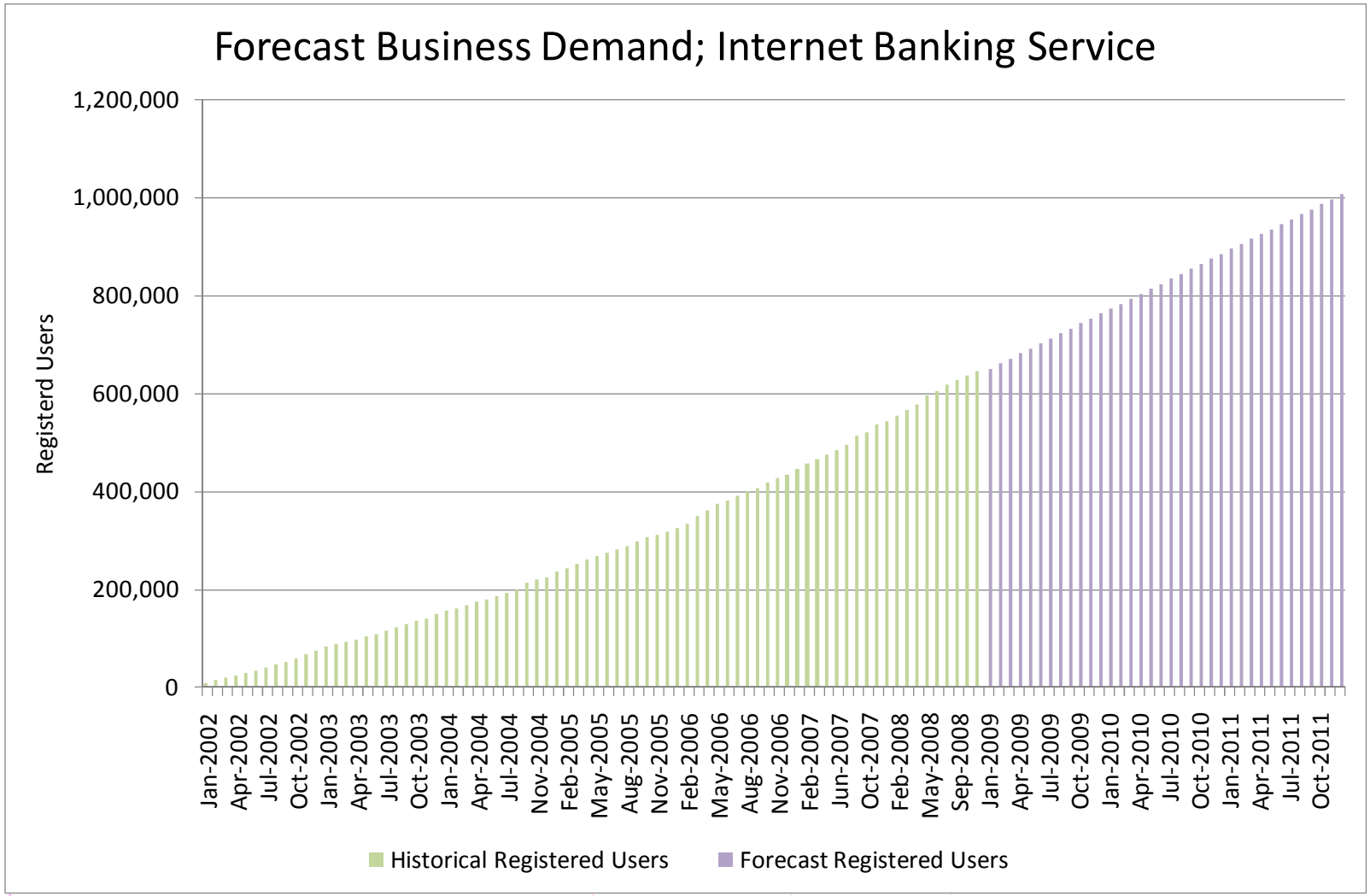
Future Business Initiatives	• Expansion of the functionality of the IT service
Business Demand Forecast	• Organic growth • Forecast demand over the planning period

Business Demand Forecast

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Business Demand Forecast

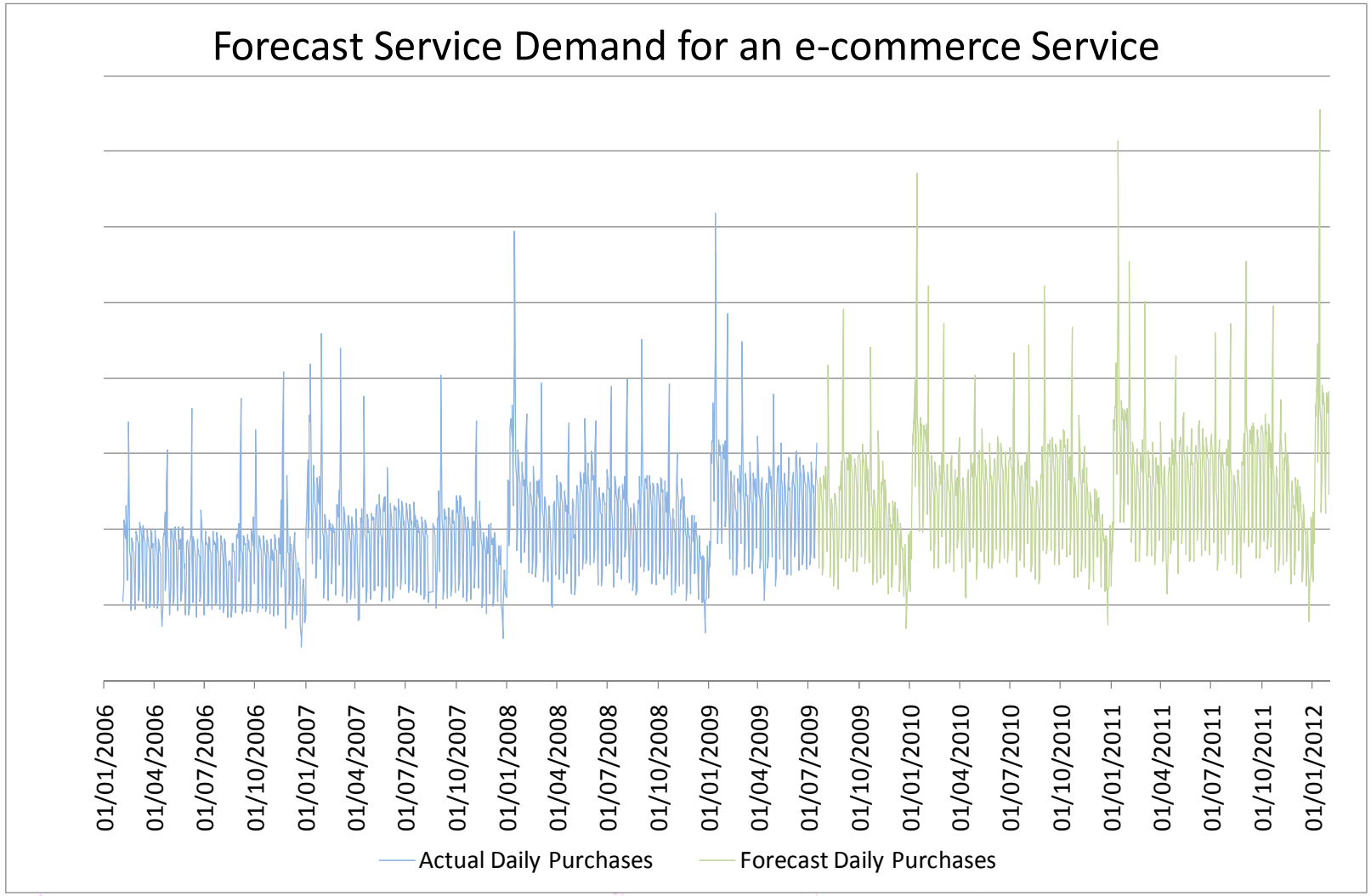


Service Capacity Management

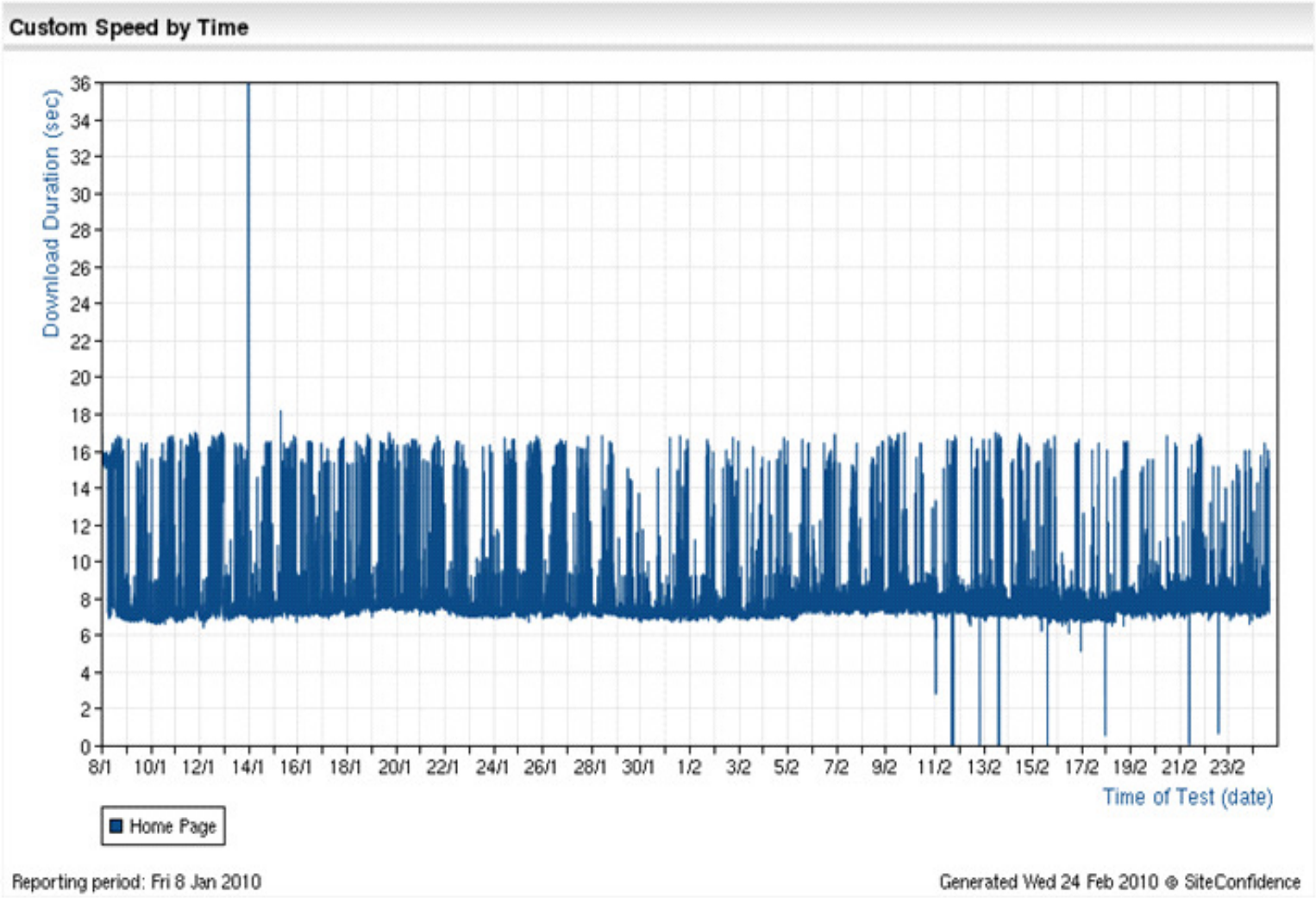
Current service performance	• Availability • Response time experienced by the end user • Service performance of third parties • Reference any service performance incidents
Service Demand Forecast	• Forecast demand over the planning period • Driven by the business demand forecast

Service Demand Forecast

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Service Performance for an e-Commerce Service



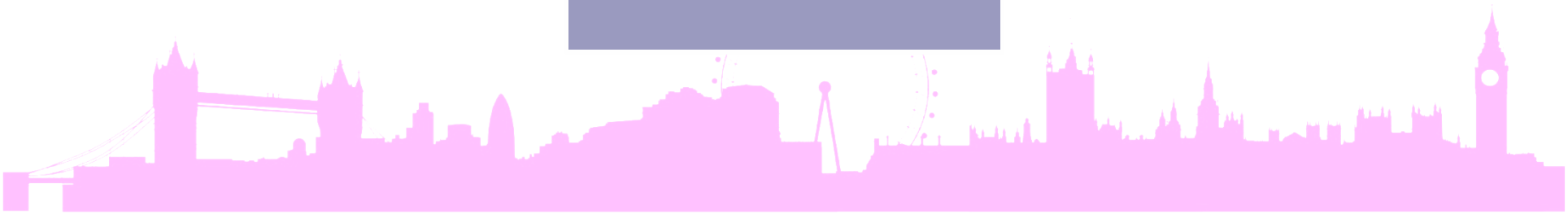
Component Capacity Management

- An assessment of the past
- Capacity plan is forward looking so avoid making this section too detailed

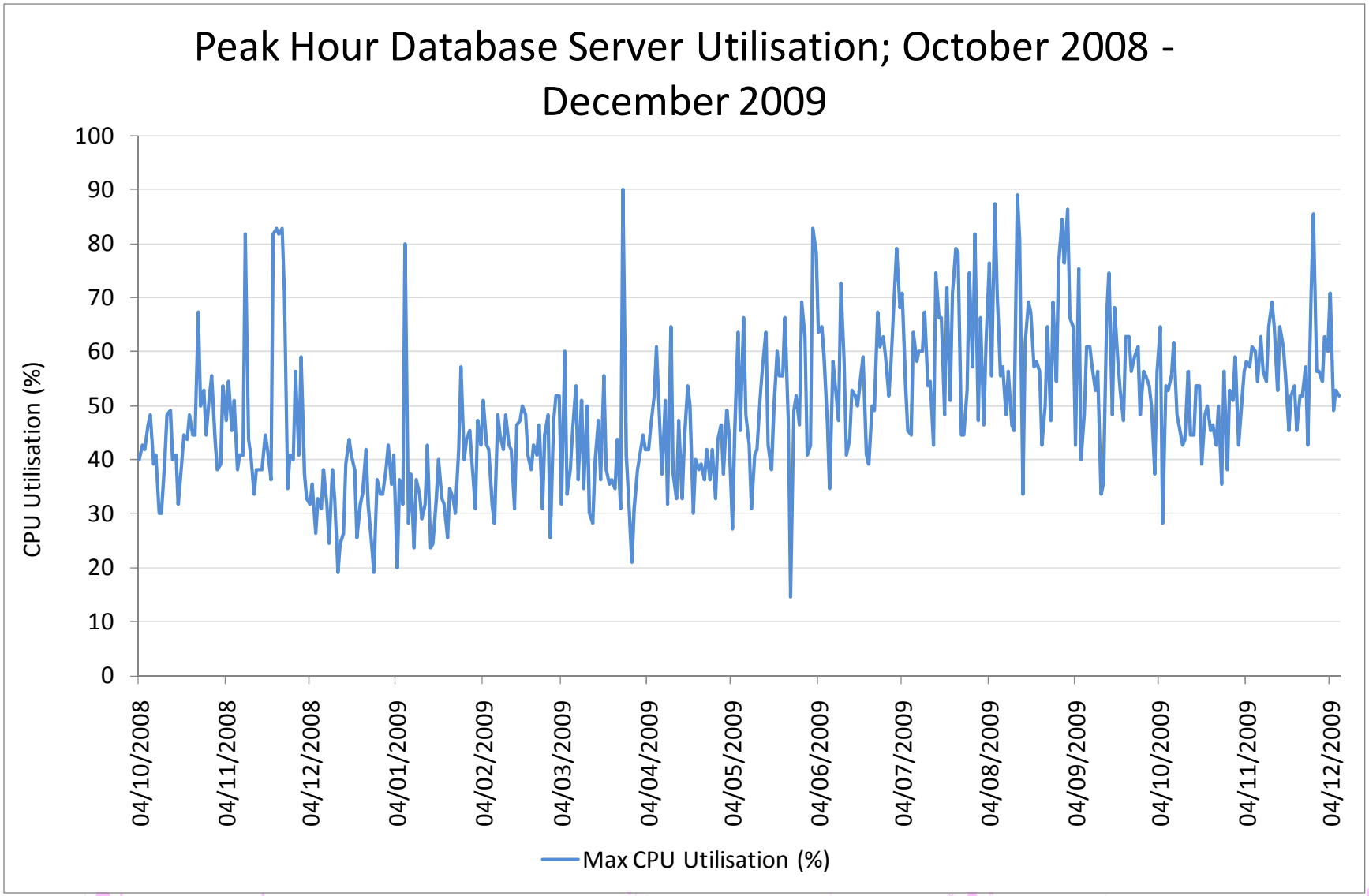
Current status

Discussion of physical component capacity constraints

Discussion of logical component capacity constraints



Component Capacity



Capacity Plan Scenarios

Business Planning Scenarios

- Merger/acquisitions
- Normal day
- Peak day
- Step change in business demand
- Addition of a new client onto the service
- What is the capacity of the service?
- Business disruption scenarios

Technical Planning Scenarios

- Technical failure scenarios
- Technical infrastructure upgrades
- Removal of a subset of servers during maintenance slots



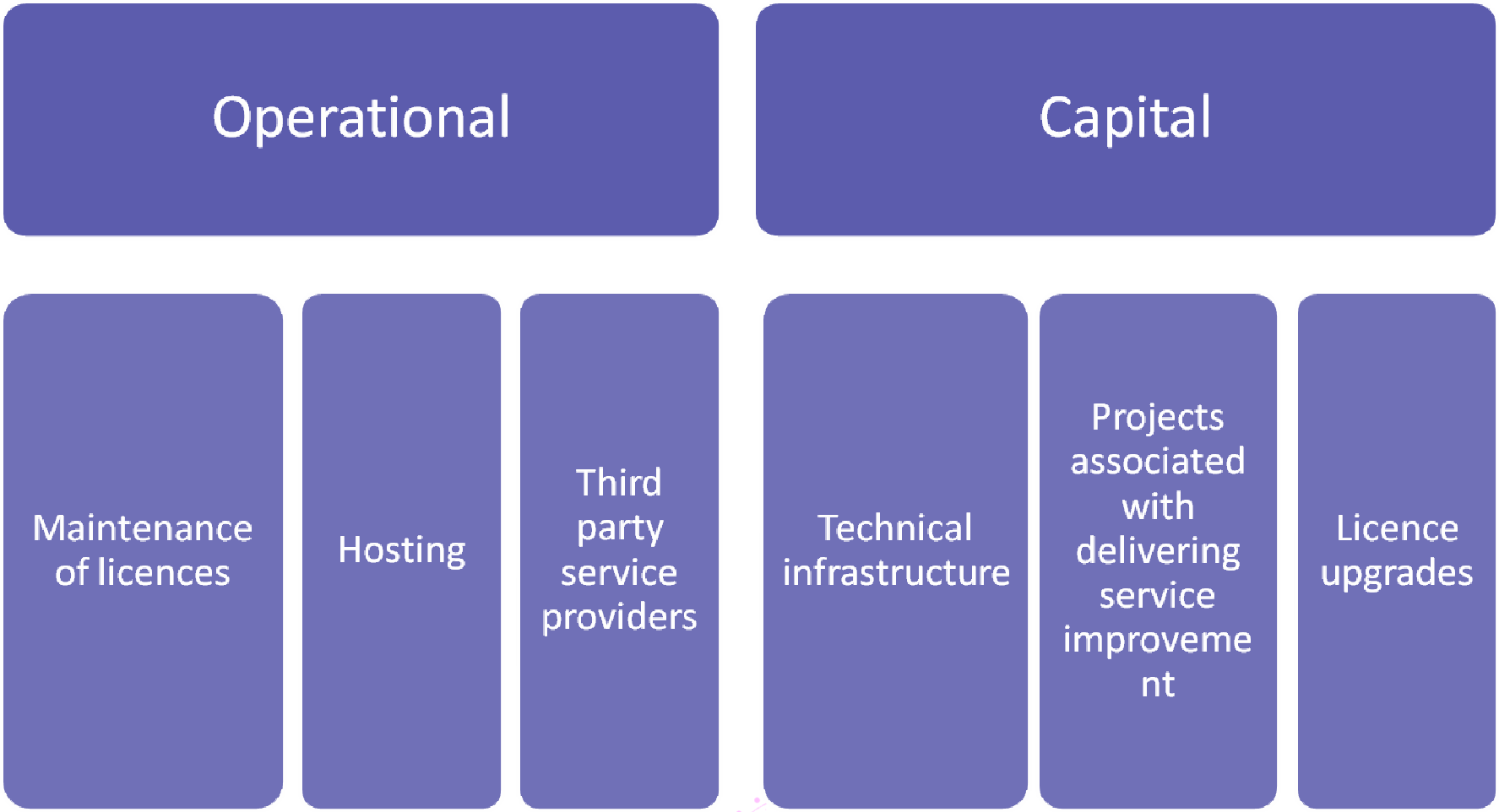
Recommendations

- **Clear recommendations on the capacity required to meet service levels**
  - Recommendations under different business/technical scenarios
- **Assign owners to recommendations after the capacity plan review meeting**
- **Assign target completion dates**

Costs

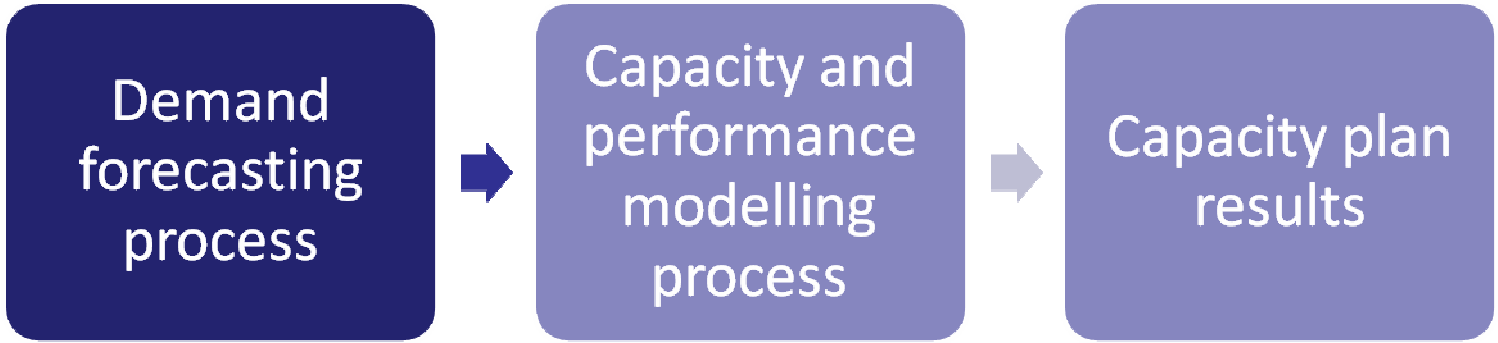
- Cross-charging between IT services that use shared technical infrastructure
- Operating costs
- Capital costs
- Relate business demand to cost

Costs

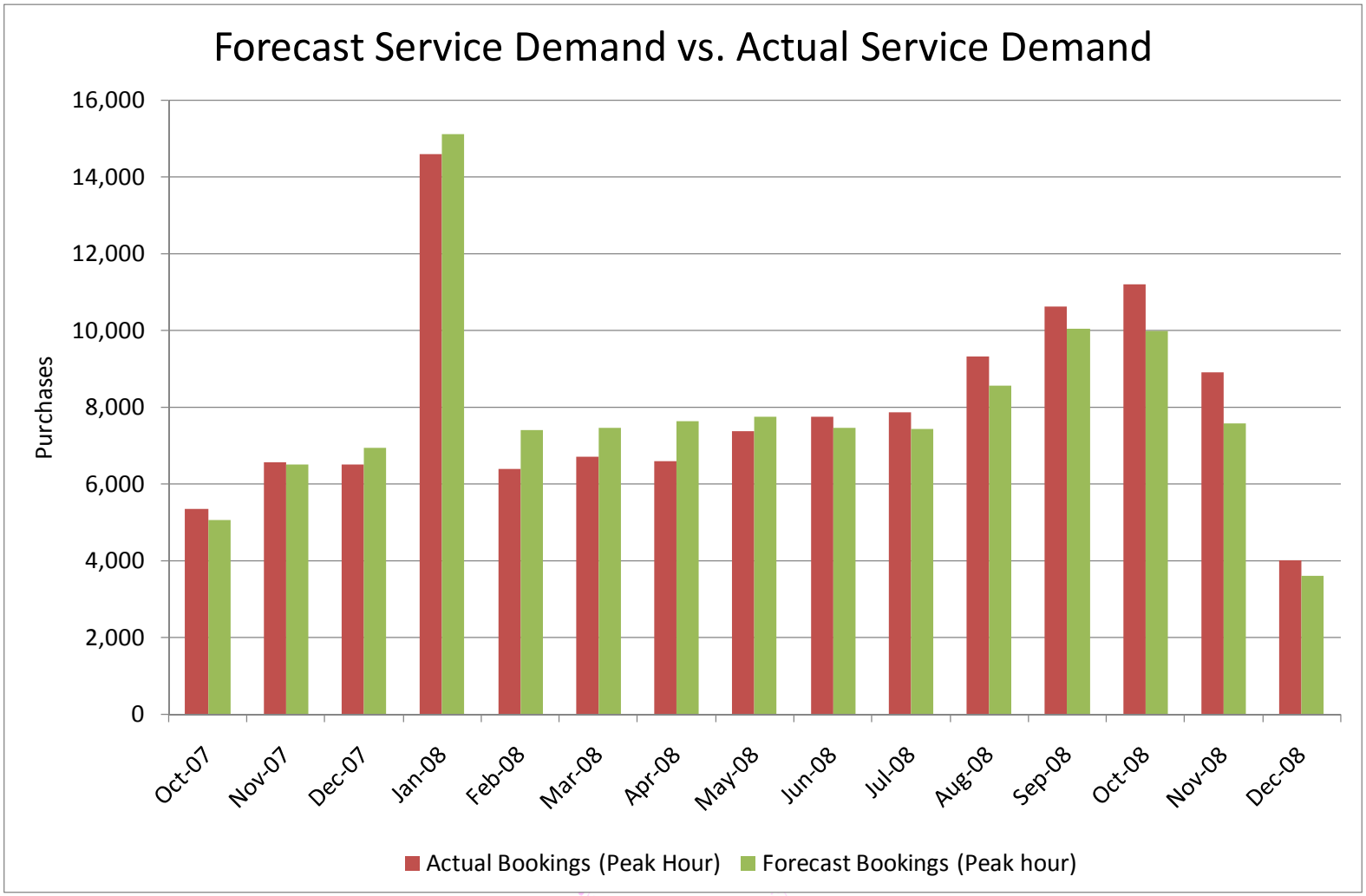

Operational
Capital
Maintenance of licences
Hosting
Third party service providers
Technical infrastructure
Projects associated with delivering service improveme nt
Licence upgrades

Plan vs. Actual

- **Two aspects here:**
  - The accuracy of the demand forecast
  - The accuracy of the capacity and performance model

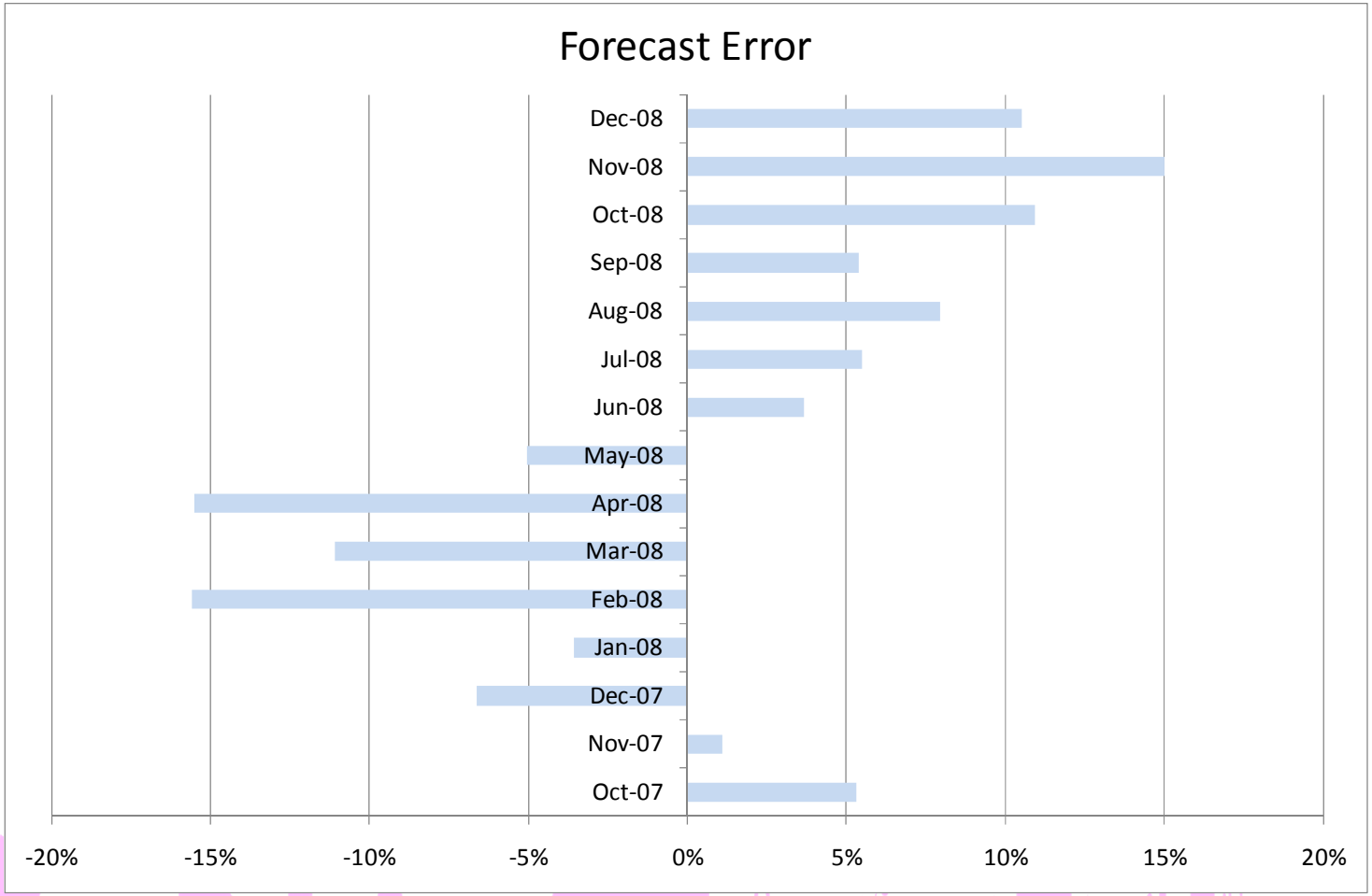


Plan vs. Actual; Demand Forecast



Plan vs. Actual; Demand Forecast

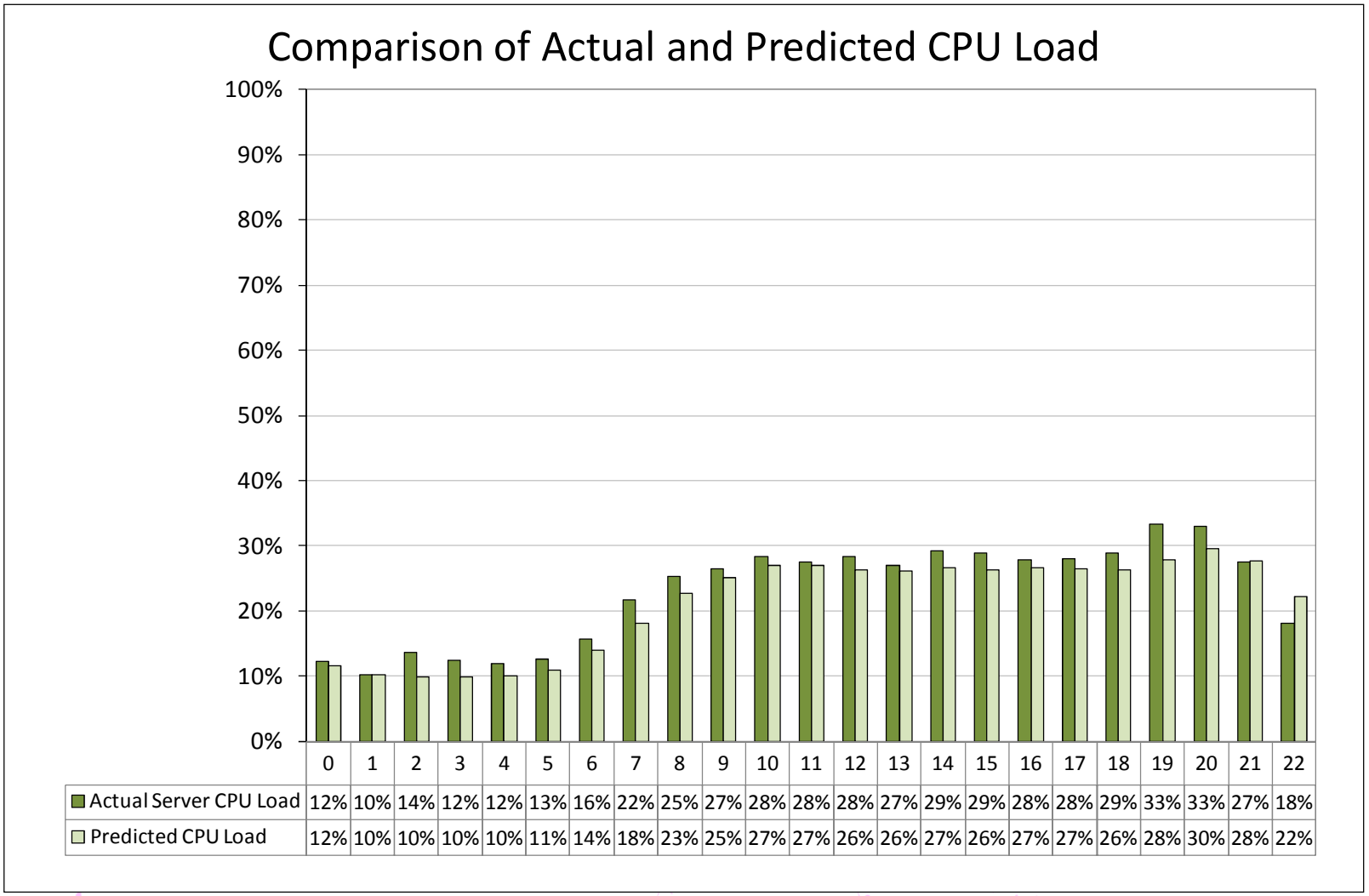
- The mean absolute percentage error (MAPE) is 12% here:



Plan vs. Actual; Capacity Model

- For a given level of observed production demand
- Compare actual capacity utilisation against modelled (predicted) capacity utilisation
- Same approach for:
  - Response time
  - Throughput

Plan vs. Actual; Capacity Model



	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22
Actual Server CPU Load	12%	10%	14%	12%	12%	13%	16%	22%	25%	27%	28%	28%	28%	27%	29%	29%	28%	28%	29%	33%	33%	27%	18%
Predicted CPU Load	12%	10%	10%	10%	10%	11%	14%	18%	23%	25%	27%	27%	26%	26%	27%	26%	27%	27%	26%	28%	30%	28%	22%

Options for Service Improvement

- **Capacity manager has a holistic view of the service**
- **Thus well placed to identify improvements**
- **Examples from client engagements:**
  - Application tuning to reduce capacity requirements
  - Improve third party performance to improve overall service performance

Appendices

- **Demand forecast methodology**
- **Capacity and performance modelling methodology**
  - Analytical
  - Simulation modelling approach
- **Methodology for plan vs. actual**

Tackling Inertia

It is vital that a face to face meeting is held with stakeholders to explain key points of the plan and agree action points

Track status of previous Capacity Plan recommendations in an Action Log

Conveying the Message

- In our experience Capacity Plans for complex IT services can be up to 60 pages long
- The target audience of the capacity plan is varied
- The plan in document format provides a detailed record
- A presentation of a few slides should be used to communicate to key stakeholders:
  - Senior IT management
  - Service owners
  - Finance



Conveying the Message

- **Example capacity plan summary for an online retailer:**

Scenario name	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10
Peak Period	Peak hour Dec 2011	Peak hour Dec 2011	Peak hour Dec 2011	Peak hour Dec 2012	Peak hour Dec 2013	Peak hour Dec 2011	Peak Hour Dec 2011	Peak Hour Dec 2011	The capacity of the service	The capacity of the service
Data centre failure scenario?	×	✓	✓	✓	✓	✓	✓	✓	✓	✓
Outstanding optimisations of the application	×	×	✓	✓	✓	×	×	✓	×	✓
Migration of all partners to the B2B service	×	×	×	×	×	✓	×	×	×	×
Business disruption scenario	×	×	×	×	×	×	✓	×	×	×
Step change in demand	×	×	×	×	×	×	×	✓	×	×
Capacity of the service (number of purchases)									185,000	241,000

Conveying the Message

- **State what the capacity of the service in units of business demand**

Industry	IT Service	Business demand unit
Airline	Flight scheduling system	Number of aircraft
Retail	Stock management system	Number of stores
Mobile phone operator	Billing system	Number of subscribers
Finance	Real-time stock market trading system	Number of trades
Government	Benefits payment system	Number of benefits recipients
Retail	E-commerce	Number of sales



Frequency of Capacity Plan

"The production and update of a Capacity Plan should occur at pre-defined intervals. It is essentially an investment plan and should therefore be published annually, in line with the business or budget lifecycle, and completed before the start of negotiations on future budgets. A quarterly re-issue of updated plan may be necessary to take into account changes in business plans, to report on the accuracy of forecasts and to make or refine recommendations."

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Frequency in Different Problem Domains

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- Different services require different frequencies
- Some examples from client engagements:

Problem Domain	Characteristics	Capacity Plan Frequency
Start-up network services provider	Rapid growth Frequent technological change Unreliable business demand forecasts	Monthly
Retail bank, customer account management	Steady growth Mature service	Annual
e-commerce service	Rapid growth	Quarterly

Summary

- Capacity plan is the chief deliverable of the Capacity Management process
- Should be forward looking rather than retrospective
- We have addressed the essential contents of any capacity plan
- Capacity plan frequency should be driven by the characteristics of the business
- Important to the convey the message using capacity plan summaries